

Year 12 Physics
Sem. 1 Exam 2016

Format

Reading Time: 10 mins

Working Time: 3 hrs

Structure

Section 1: 14 short-answer questions

Section 2: 5 problem-solving questions

Section 3: 2 reading comprehensions - 14 questions

Areas of Study

Waves

- Sound versus light waves.
- Wave graphs - amplitude, wavelength, period, particle movement.
- $v = f\lambda$

Motion

- Vectors - one and two-dimensional, components.
- Projectile motion - vertical and horizontal movement.
- Circular motion - vertical, horizontal, conical pendulum, banking.
- Satellites - orbital speed, period, geo-stationary orbit, energy.

Electricity

- Electric and magnetic field drawings.
- Force on current-carrying wires, isolated charges.
- Motors - calculations, torque, explanation of construction.
- Induced EMF - straight conductors, coils.
- Generators - calculations, explanation of construction.
- Transformers - calculations, construction, operation.
- Transmission of electricity - power grid.